

Shivam Bhardwaj

+1-605-592-1693

[linkedin.com/in/shivam-kumar-bhardwaj/](https://www.linkedin.com/in/shivam-kumar-bhardwaj/)

github.com/bshivamkumar

Education

Maharishi Markandeshwar College of Engineering

Bachelor of Technology in Computer Science (CGPA: 8.34 / 10.00)

08/2014 - 05/2018

Ambala, Haryana

- **Coursework:** Data Structures and Algorithms Python, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks

South Dakota State University

MS in Data Science (GPA: 3.2)

08/2024 - 12/2025

Brookings, South Dakota

Experience

Research & Academic Experience

Lab Assistant, SDSU

Sept 2024 - Present

- Analyzed **large plant-genome datasets** to train and test predictive ML models.
- **Developed modular, reproducible pipelines** for automated data preprocessing and validation.
- Applied **CI/CD-style automation** to ensure code versioning, reproducibility, and scalability.
- Experience directly relevant to **software maintenance, workflow automation, and repository evolution analysis**.

Atos Global

Software Engineer

Oct 2018 – Jul 2021

Bengaluru, Karnataka

- Achieved **99.8% uptime** for **data pipelines** managing **streaming and transactional data** from **8 primary sources** using **Apache Spark, AWS Redshift, and Python**
- Automated **ETL workflows** and **CI/CD deployment pipelines** using **Jenkins**, reducing **manual workloads by 29%** and **deployment time by 30%**.
- Integrated **SAP, Collibra, and Salesforce** data sources using **Python APIs**, delivering **Tableau-compatible views** that enhanced **business intelligence and reporting capabilities**.
- Implemented **Prometric feature enhancements**, cutting **assessment costs by 22%** while maintaining robust **system reliability**.
- **Tech Stack:** React | MySQL | Python | Spark | AWS Redshift | Jenkins | Tableau | SAP | Collibra | Salesforce | Agile | API Integration | Data Engineering.

Cognizant

Data Engineer

Aug 2021 – Feb 2023

Bengaluru, Karnataka

- Built and optimized **data pipelines (~10 TB/day)** using **Apache Spark** and **AWS Glue**, improving reliability by **35%**.
- Migrated **20+ ETL workflows** from **Oracle** to **AWS Redshift**, cutting costs by **30%** and boosting scalability.
- Deployed **real-time streaming** with **Apache Kafka** for **fraud detection**, reducing latency from **15 min to <1 min**.
- Automated **Airflow + Terraform** deployments, lowering errors and deployment time by **40%**.
- Maintained **99.8% pipeline uptime** across **sources**, ensuring continuous data availability.
- **Tech Stack:** Apache Spark | AWS Glue | AWS Redshift | Apache Kafka | Airflow | Terraform | Python | SQL | Oracle | AWS

Capgemini

Sr. Data Engineer

Mar 2023 – Aug 2024

Bengaluru, Karnataka

- Migrated **Oracle databases** to **AWS Redshift** via **Athena** and **S3**, saving **\$678K annually** and improving **performance by 14%**.
- Designed **real-time data pipelines** handling **150M+ semi-structured records** from **30+ sources** using **Apache Spark** and **Kafka**.
- Architected **scalable data pipelines** supporting **125K+ daily active users** for new product launches.
- Standardized **data dictionaries** across domains, enhancing **analytics consistency and usability**.
- **Tech Stack:** AWS Redshift | Athena | S3 | Apache Spark | Apache Kafka | Python | SQL | Data Modeling | ETL | Cloud Data Architecture

Projects

bactofunc | NextFlow, R, Python, MongoDB, MySQL, Json Web Token

- **Bactofunc** is a Python package for **functional genomic analysis of bacterial genomes**, enabling **annotation, comparison, and pan-genome** exploration.
- It streamlines **variant calling, phylogenetic, and ortholog clustering** to accelerate **microbial bioinformatics** workflows.

Keystroke Dynamics  | *XGBoost, Scikit-learn, NumPy & Pandas, Matplotlib*

- This repository implements a **comparative analysis of machine learning algorithms** for **anomaly detection in keystroke dynamics biometrics**.

Technical Skills

Programming Languages: Python, R, JavaScript, Spark
Framework and Libraries: React.js, Redux, Angular, Node.js
Cloud/DevOps & Tools: Jenkins (CI/CD), AWS, Git, Postman
Databases: MySQL, MongoDB